Roll No. .....

MAI–206(DSI)

M. B. A. (AI & DS) (SECOND SEMESTER)

END SEMESTER EXAMINATION, July/Aug., 2022

DATA SCIENCE USING PYTHON

Time : Three Hours

Maximum Marks : 100

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *two* sub-questions among (a), (b) and (c) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section—A

1. (a) Explain the different string formats available in Python with examples. (CO1)

(b) Discuss the int(), float(), str(), chr() and complex() type conversion functions with examples. (CO3)

(c) Data from the file "weather.xlsx "has to be loaded into a pandas dataframe df_weather which when printed is as shown below : (CO6)

In [38] : df_weather

P. T. O.

Out [38] :

	Direction	Temperature	Windspeed	Humidity
0	East	49	10	78
1	West	54	5	80
2	North	35	8	92
3	South	42	15	70

Of the following set of statements which of them can be used to move the column "Direction" into a separate dataframe :

(i) df_weather [['Direction']]

(ii) df_weather ['Direction']

(iii) df_weather . loc [:, ['Direction']]

(iv) df_weather . iloc [:, 0]

2. (a) Discuss the following list functions : (CO2)

(i) len()

(ii) sum()

(iii) any()

(iv) all()

(v) sorted()

(b) Describe the **is** and **is not** operators and type () function. Also, discuss why Python is called as dynamic and strongly typed language. (CO3)

(c) Data from the file "brand_data.csv" has to be loaded into a pandas dataframe. A snippet of the data is shown below : (CO5)

0, 1, 2, 3

brand, type, cost, price

BR1, Clnr, 12, 15

BR2, util, 23, 34

BR3, lux, 189, 191

Br4, txtl, 150, 130

What is the right instruction to read the file into a dataframe df_brand with separate columns ?

(i) pd.read_csv ("brand_data.csv", index_col = 0, header = 1)

(ii) df_brand = pd.read_csv ("brand_data.csv", header = 1)

(iii) df_brand = pd.read_csv ("brand_data.csv", header = None)

(iv) df_brand = pd.read_table ("brand_data.csv", delimiter = ', ', header = 1)

3. (a) Compare discrete random variables with continuous random variable with example. (CO2)

(b) Describe data cleaning. Write a code to drop a column from a data set. (CO3)

(c) Consider a dictionary city created with the following keys and values. (CO4)

City = {'Delhi' :3, 'Bengaluru' : 5, 'Chennai' : 4, 'Kolkata' : 6, 'Mumbai' : 7}

Through which all possible way/ways can we access the value 5 from the dictionary city ?

1. city ['Bengaluru']

2. city.get ['Bengaluru']

3. city.values()
4. list(city.values( ))[1]
5. None of the above

4. (a) What is Unstructured data? How to analyze unstructured data ? (CO1)

(b) Differentiate between lists and dictionaries in Python. (CO2)

(c) Explain import pandas as pd, import numpy as np. (CO4)

Count the number of elements in the below list.

list_lens = [["October", 24, ["2021"]]]

(a) 2

(b) 1

(c) 3

(d) 0

(e) None of the above

Section–B

5. **Case Study :** (CO6)

A Government company delivers subsidies to individuals based on their income. This company obtained a large set of authenticated data on individual income, demographic parameters, and a few financial parameters.

This company wishes us to develop an income classifier for individuals. Take any hypothetical dataset. Write the coding for Classification, Prediction and Confusion Matrix.